

Leonard Wong

AI & Platform Software Engineer | Agentic Systems | Secure Software Delivery

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PROFESSIONAL SUMMARY

Hybrid AI and platform software engineer with 4+ years delivering public-sector software at NCS Group, backed by earlier work in high-volume data systems, API integration, and application security. Builds governed agentic systems, LLM-enabled workflows, and developer tools using TypeScript/Next.js/PostgreSQL, Swift/MCP, C#/ASP.NET, and cloud/data platforms. Combines policy-as-code, human-in-the-loop approvals, security-by-design, UAT, incident management, and stakeholder enablement. Published ITICSE researcher.

TECHNICAL SKILLS

- AI & Agentic Systems:** Built governed LLM applications and agentic workflows with RAG, prompt engineering, tool/function calling, MCP/JSON-RPC, responsible/explainable AI, and human approval controls.
- Agentic AI Tools:** Used Claude Code, Codex, and Kiro CLI to create reusable skills/prompts, tailor tool-specific instructions, validate outputs, and standardise task handoffs.
- Software Engineering:** Built APIs, dashboards, workers, macOS CLI/GUI tools, and data pipelines with TypeScript/Node.js/Next.js/Fastify/React, Swift/SwiftUI/EventKit, C#/.NET, Python, SQL, and web technologies.
- Data & Platforms:** Used PostgreSQL/Prisma and SQLite for persistence, Redis/BullMQ for durable jobs, DynamoDB/ETL for large-scale data conversion, Docker for local services, and GitHub Apps/OAuth/webhooks for integrations.
- Security & Quality:** Applied policy-as-code, signed webhooks, fail-closed configuration, secret redaction, AES-GCM/PBKDF2, OWASP testing, SAST/DAST, CodeQL, Vitest, Playwright, and CI/CD to secure and validate releases.
- Delivery & Product:** Led UAT, incident resolution, user training, stakeholder validation, Agile delivery, and platform/product/service design from requirements through support.

PROFESSIONAL EXPERIENCE

NCS Pte. Ltd. | Singapore

Jan 2022 - Present

AI Software Engineer

Apr 2026 - Present

- Build LLM applications and agentic workflows using RAG to ground responses in approved data, function calling to invoke application tools, and prompt engineering to define repeatable task flows.
- Apply responsible and explainable AI, human review, software-security controls, and operational-readiness practices to support safe delivery and operation.
- Translate stakeholder needs into testable requirements; coordinate validation, user enablement, and post-release support.

Software Engineer

Jan 2022 - Mar 2026

- Owned the full delivery lifecycle of a confidential internal system, from implementation and deployment through post-release support.
- Planned and facilitated UAT by authoring test scripts, coordinating stakeholder sessions, and driving issue resolution.
- Managed operational incidents and delivered user training and ongoing support to sustain adoption and service continuity.

Leadership: NCS Young Leaders; Brand Ambassador

SELECTED PUBLIC ENGINEERING PROJECTS

AgentForge Merge Guard

2026

TypeScript | Fastify | Next.js | PostgreSQL/Prisma | Redis/BullMQ | GitHub Apps

- Built a GitHub-first PR governance service with deterministic detectors, YAML policy-as-code, required evidence, reviewer routing, check runs, and durable Change Control Records.
- Implemented an API/dashboard/worker monorepo with signed webhooks, fail-closed production configuration, metadata-only storage, and secret redaction; optional AI remains advisory and disabled by default.

Agentic Control Plane

2026

TypeScript | Next.js | PostgreSQL | Background Workers | Integrations

- Built a modular control plane for goals, commitments, approvals, automations, memory, documents, and integrations around a 6-stage bounded execution loop with durable jobs, risk classification, and higher-risk approval gates.

Agent Capability Prompt CLI

2026

Node.js | Claude Code | Codex | Kiro CLI | Prompt Engineering

- Built a zero-dependency Node.js CLI with 8 presets for Claude Code, Codex, Kiro CLI, and other agents; generates reusable skills/prompts, injects task examples, applies tool-specific guidance, and requires validation and structured handoff.

Apple Calendar MCP & CLI

2026

Swift | SwiftUI | EventKit | MCP/JSON-RPC | Homebrew

- Built a local macOS tool exposing 5 MCP operations - list, search, create, update, and delete - plus CLI/GUI interfaces, JSON/table output, read-only mode, and calendar write allowlists.

PassSync

2026

Swift | SwiftUI | 1Password CLI | macOS Keychain | SQLite | AES-GCM

- Built a security-sensitive credential comparison and sync tool with dry-run defaults, encrypted pre-mutation backups, fail-closed conflict handling, passkey/TOTP guardrails, restore verification, and hash-chained audit receipts.

Governed Portfolio Delivery

2020 - 2026

Node.js | GitHub Actions | CSP | Playwright | CodeQL | Static Generation

- Generated portfolio and resume pages from validated JSON and enforced a 14-step release gate spanning security, content parity, performance, telemetry, links, dependencies, vendor governance, integration, and accessibility.

EARLIER EXPERIENCE

Singapore Management University <i>Research Assistant</i>	Dec 2021
• Evaluated 88 IS216 projects using OWASP ZAP, commercial DAST tools, and static analysis; findings supported peer-reviewed research on integrating security into web programming education. • Co-authored the ITiCSE 2022 paper "XSS for the Masses," nominated for the conference Best Paper Award.	
Alpha Ori Technologies <i>Maritime Tech Solutions Intern</i>	May - Jul 2020
• Automated conversion of 150M+ rows containing 3B+ data points into AWS DynamoDB using C#/.NET, JSON, and ETL workflows. • Built API proof-of-concepts for 3 vendors and implemented data sorting and voyage analysis to evaluate technical fit and operational trends.	
OSM Thome (formerly Thome Ship Management) <i>IT Intern</i>	May - Oct 2018
• Built an analytics portal presenting 15 decision data points, a ship-agency workflow supporting 160+ emails/hour, and API enrichments covering 230+ vessels.	
Rightspot Intern - Resolved 20+ bugs in 2 weeks; built applicant-status and job-listing dashboards covering 250 applicants and 500+ companies.	Jul - Oct 2016
SamMobile Contributing Writer - Published 1-2 articles per week for a 1.5M+ member audience, covering mobile software updates and product insights.	Jul - Dec 2013

EDUCATION

Singapore Management University - BSc Information Systems (Cyber Security)	2017 - 2022
Nanyang Polytechnic - Diploma in Business Informatics with Merit	2014 - 2017
Gachon University - Overseas Exchange	2015

PUBLICATION

XSS for the Masses: Integrating Security in a Web Programming Course using a Security Scanner - ITiCSE 2022; co-author; nominated for Best Paper Award.	
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CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

AI Singapore - AlxTech (Certificate of Completion)	2026
AI Singapore - Certificate of Completion: LLM Application Developer Programme	Feb 2026
NUS-ISS - AI and Cybersecurity	Jun 2025
NUS-ISS - Explainable and Responsible Artificial Intelligence	May 2025
NUS-ISS - Graduate Certificate in Designing and Managing Products and Platforms	Mar 2025
NUS-ISS - Service Design	Jun 2024
NUS-ISS - Architecting Platforms as a Business	May 2024
NUS-ISS - Digital Product Strategy	May 2024
Singapore Computer Society - Certification in Managing Digital Products	Mar 2024
NUS-ISS - Managing Digital Products	Mar 2024
NUS-ISS - Graduate Certificate of Architecting Scalable Systems	Feb 2024
NUS-ISS - Platform Engineering	Dec 2022
NUS-ISS - Cloud Native Solution Design	Nov 2022
NUS-ISS - Architecting Software Solutions	Aug 2022
NUS-ISS - DevOps Engineering and Automation	Mar 2022
Scrum.org - Professional Scrum Master I (PSM I)	Feb 2022

HONORS, LEADERSHIP & LANGUAGES

• MPA Global Internship Award (2019); Quantedge Foundation Scholarship (2017); Nanyang Polytechnic Director's List for 5 semesters (top 15%). • Central Singapore CDC mentor/project co-leader (2018-2020); SMU Apolitical Society Honorary Financial Secretary; organized a ministerial dialogue with 90+ attendees. • Agentic AI practice: use Claude Code, Codex, and Kiro CLI to create and validate reusable skills, prompts, and handoff workflows; authored the Human Operating Model Codex skill and an 8-preset cross-agent CLI. • Languages: English and Chinese - native or bilingual proficiency.	
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